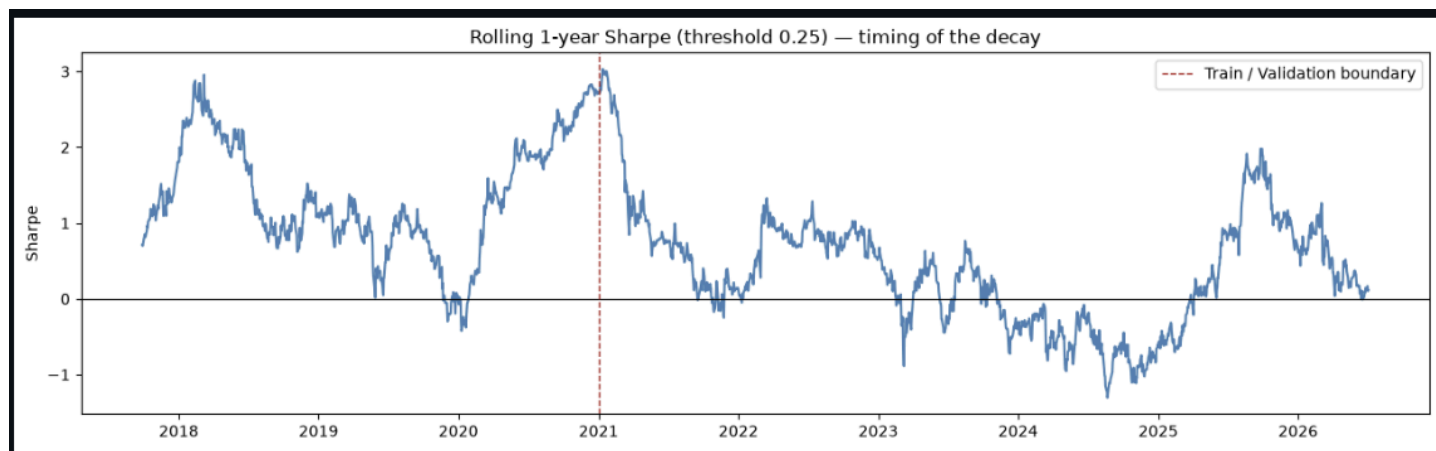


## Factor A and Crude Oil

Factor A carries genuine but decaying predictive information about oil. It supported a strong strategy through 2020 (Sharpe 1.36), deteriorated sharply at the start of 2021, and has since stabilised at a weak residual (Sharpe 0.32 out of sample). Not deployable as a standalone at current strength; retains value as an uncorrelated portfolio component.



## The rules

- Signal: Factor's 5-day momentum, standardised by a 60-day rolling z-score.
- Position: Long if  $z > +0.25$ , short if  $z < -0.25$ , otherwise flat
- Sizing: Scaled to a 10% annualised volatility target from oil's trailing 60-day volatility
- Frequency: Avg ~11-day holding period, ~25x annual turnover, assume costs charged at 5bps per trade

## Results

| Period                   | Sharpe | Annual Return | Annual Vol | Max drawdown | Oil (passive) |
|--------------------------|--------|---------------|------------|--------------|---------------|
| Train (2016-20)          | 1.36   | 14.4%         | 10.6%      | -10.6%       | 0.28          |
| Validation (2021-23)     | 0.23   | 2.3%          | 10.0%      | -11.3%       | 0.53          |
| Test 2024-26 (Untouched) | 0.32   | 3.3%          | 10.1%      | -10.9%       | 0.16          |

## Methodologies

- Momentum, not level: The factor's return predicts forward oil returns (rank IC +0.065 at 5 days,  $p = 0.001$ ). Its level appeared to predict (IC +0.063) but sat above zero on 67% of days (long twice as often as short), simply because the factor trends upwards, so its apparent edge was really just a bet on oil rising. Rebuilt a different way, the signal vanished ( $p = 0.19$ ). It was rejected as disguised oil beta.
- 5-day horizon: Predictive content peaks at 5 days and is insignificant by 20, which sets the rebalancing frequency.
- Volatility targeting: Oil's realised volatility ranged from ~20% to over 100%. Holding a fixed position size means unknowingly taking five times more risk in a crisis than in a quiet market. Instead, when oil gets volatile, the position automatically shrinks.
- Both 20 and 21 April 2020 were excluded: At  $-\$37.63$ , percentage returns break down: the 20th computes as  $-306\%$ , and a short position across it would have booked a spurious 400% gain. The 21st is corrupted too, its return divides by that negative price, giving  $-126.6\%$  despite the price rising \$47.
- COVID Dependency check: Every standardisation window from 30 to 120 days shows the same strong-then-weak pattern. Excluding COVID entirely, the training Sharpe only falls from 1.36 to 1.20.

## Recommendation

1. Don't trade Factor A alone. A net Sharpe of  $\sim 0.3$ , with transaction costs consuming roughly a third of the gross edge, does not justify capital against realistic alternatives.
2. Worth explaining the factor decay. The drop happened suddenly in early 2021, not gradually. Something changed. Finding out the reason, whether the signal got crowded, the oil market changed, or whatever drives the factor shifted, would tell you if the edge is gone for good or could come back.

## Limitation

1. Single factor, single asset: The signal is read as weakly positive, not as a precise estimate. The leftover edge isn't steadily weak, 2024 was bad, and 2025 was good.